Stats 102B Homework 1

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(1) Angle Between Two Vectors

(a) get_angle() Counterclockwise angle between two vectors

```
get_angle = function(x, y) {
  # Calculates the (counterclockwise) angle between two 2-dimensional vectors

  # Calculate angle
  radians <- acos(dot(x,y) / (Norm(x) * Norm(y)))
  degrees <- radians * (180/pi)

  # Make angle counterclockwise
  cross_prod <- x[1]*y[2] - x[2]*y[1]

  if(cross_prod > 0) {
    degrees <- 360 - degrees
  }

  degrees
}
```

(b) Vector pairs giving 17, 110, and 270 degrees

```
get_angle(c(cos(17 * (pi/180)),sin(17 * (pi/180))), c(1,0))
```

```
## [1] 17
```

```
get_angle(c(cos(110 * (pi/180)),sin(110 * (pi/180))), c(1,0))
```

```
## [1] 110
```

```
get_angle(c(cos(270 * (pi/180)),sin(270 * (pi/180))), c(1,0))
```

```
## [1] 270
```

(2) Matrix Inverse Properties

Given that A^{-1} and B^{-1} exist:

(a) Show that $(A^T)^{-1} = (A^{-1})^T$

$$AA^{-1} = I$$
$$(AA^{-1})^T = I^T$$
$$A^T(A^{-1})^T = I$$
$$(A^T)^{-1}A^T(A^{-1})^T = (A^T)^{-1}I$$
$$(A^{-1})^T = (A^T)^{-1}$$

(b) Show that $(AB)^{-1} = B^{-1}A^{-1}$

$$I = AA^{-1}$$
$$I = AIA^{-1}$$
$$I = A(BB^{-1})A^{-1}$$
$$I = (AB)(B^{-1}A^{-1})$$
$$(AB)(AB)^{-1} = (AB)(B^{-1}A^{-1})$$
$$(AB)^{-1}(AB)(AB)^{-1} = (AB)^{-1}(AB)(B^{-1}A^{-1})$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

(3) Eigenvalue Property

A is kxk matrix and P is kxk orthogonal matrix. Show that $P^TAP and A have the same eigenvalues$

(4) Ellipses

(5) Orthogonal Projection

(6) Modelling